



BAULKHAM HILLS HIGH SCHOOL

2024

HSC Assessment Task 1

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using non-erasable black pen.
- Calculators approved by NESA may be used.
- Marks may be deducted for careless or badly arranged work.
- The Mathematics Reference Sheet is provided at the back of this paper.
- Answer all questions on the appropriate page of your answer booklet.
- For questions in Section II, show relevant mathematical reasoning and/or calculations.
- If extra paper is used, it must be clearly labelled with your NESA number and the question being answered, and stapled inside your answer booklet.

Total marks – 39

Exam consists of 7 pages.

Section I: Pages 2 - 4

5 marks

Attempt Questions 1 – 5

Allow about 8 minutes for this section

Section II: Pages 5 - 7

34 marks

Attempt Questions 6 – 8

Allow about 52 minutes for this section

Section I:**Marks****5 marks****Attempt Questions 1 - 5****Use the multiple-choice answer sheet in your booklet.**

1. Which of the following is equivalent to $\sqrt{i^3}$? **1**

(A) $e^{\frac{i\pi}{4}}$

(B) $e^{\frac{i\pi}{2}}$

(C) $e^{\frac{i3\pi}{4}}$

(D) $e^{\frac{-i\pi}{2}}$

2. The polynomial $P(z)$ has real coefficients, and $z = 3 + i$ is a root of $P(z)$. Which of the following quadratic polynomials must be a factor of $P(z)$? **1**

(A) $z^2 - 6z + 10$

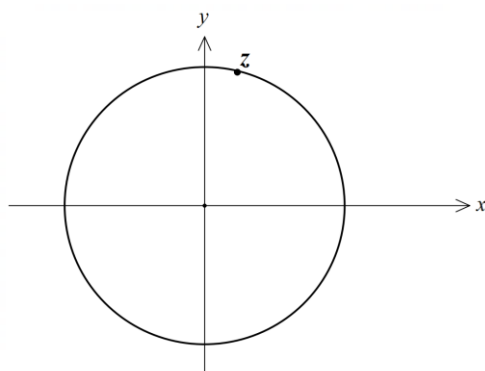
(B) $z^2 + 6z + 10$

(C) $z^2 - 6z + 8$

(D) $z^2 + 6z + 8$

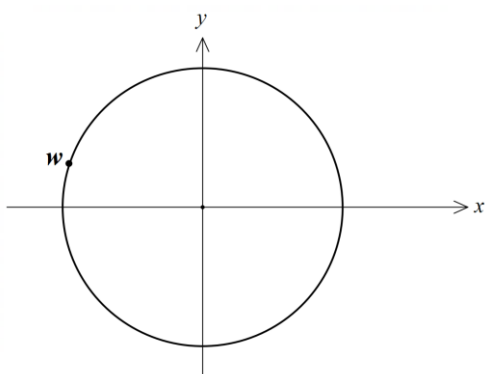
3. The Argand diagram below shows the complex number location for the complex number z .

1

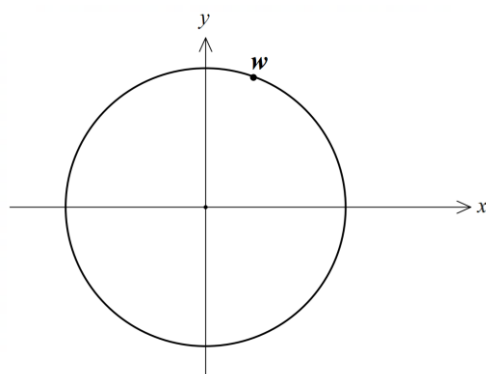


Which Argand diagram best represents $w = \frac{z^2}{i|z|}$?

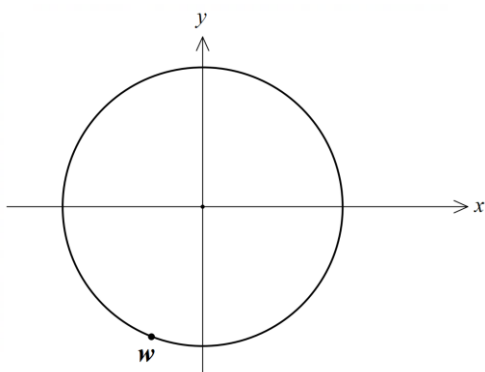
(A)



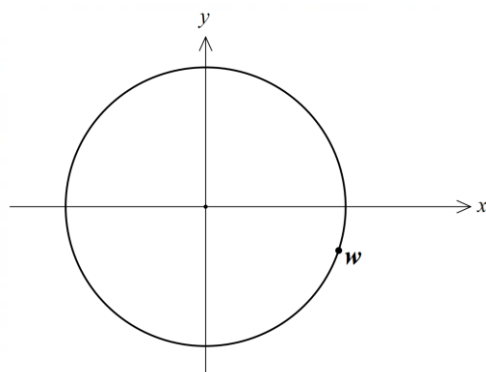
(B)



(C)



(D)



4. Which of the following expression is equivalent to $(-1+i)^n$ where n is a positive integer?

1

(A) $(\sqrt{2})^n \left(\cos \frac{n\pi}{4} + i \sin \frac{n\pi}{4} \right)$

(B) $(\sqrt{2})^{-n} \left(\cos \frac{-n\pi}{4} + i \sin \frac{-n\pi}{4} \right)$

(C) $(\sqrt{2})^n \left(\cos \frac{3n\pi}{4} + i \sin \frac{3n\pi}{4} \right)$

(D) $(\sqrt{2})^{-n} \left(\cos \frac{3n\pi}{4} + i \sin \frac{3n\pi}{4} \right)$

5. If $z^2 + z + 1 = 0$, where z is complex number then what is the value of

1

$$\left(z + \frac{1}{z} \right)^2 + \left(z^2 + \frac{1}{z^2} \right)^2 + \dots + \left(z^6 + \frac{1}{z^6} \right)^2 ?$$

(A) 12

(B) 10

(C) 14

(D) 8

End of Section I

Section II: 34 marks

Attempt Questions 6 - 8

Answer each question on the appropriate page of the answer booklet, showing all necessary working

Question 6 (11 marks)

- a) Let $w = 1 + iz$, where $w, z \in \mathbb{C}$. Express each of the following in the form of $x + iy$.
- (i) w if $z = 2 + i$ 1
- (ii) z if $w = 2 - i$ 1
- b) Given that $z = (b + i)^2$, where b is real and positive. Find the value of b if $\arg(z) = \frac{\pi}{3}$. 3
- c) Solve the equation $z^2 - 5z + (7 + i) = 0$. 3
- d) (i) On an Argand diagram shade in the region determined by the inequalities 2
- $$|z - \bar{z}| \leq 4 \text{ and } \frac{7\pi}{6} \leq \arg(z) \leq \frac{5\pi}{4}.$$
- (ii) Let z_0 be the complex number of maximum modulus satisfying the above 1
- inequalities. Express z_0 in the form of $x + iy$.

End of Question 6

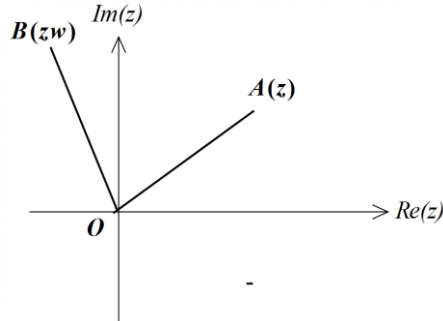
Question 7 (12 marks)

- a) Consider the quartic equation $z^4 + 4z^3 + 8z^2 + 80z + 400 = 0$, where $z \in \mathbb{C}$. Two of the roots of this equation are $(a + ib)$ and $(b + ia)$, where $a, b \in \mathbb{Z}$. Find the possible values of a . 3
- b) Sketch the locus of the point z on an Argand diagram, given $\arg\left(\frac{z-2}{z-2+4i}\right) = \frac{\pi}{2}$. 2
- c) Prove that if $|z| = 1$, then $\frac{z-1}{z+1}$ is a purely imaginary number. 2
- d) (i) Show that $z^5 + 1 = (z+1)(z^4 - z^3 + z^2 - z + 1)$. 1
- (ii) If z is a non-real root of $z^5 + 1 = 0$, prove that $z^4 + z^2 + 1 = z^3 + z$. 1
- (iii) Hence or otherwise, show that $\cos\frac{\pi}{5} + \cos\frac{3\pi}{5} = \frac{1}{2}$. 3

End of Question 7

Question 8 (11 marks)

- a) On the Argand diagram below, the complex numbers z and zw are represented by the points A and B respectively, where $z = re^{i\theta}$ and $w = e^{\frac{i\pi}{3}}$, and $r > 0$.



- (i) Explain why OAB is an equilateral triangle. 2
- (ii) Write the complex number $(z - zw)$ in exponential form in terms of r and θ . 2
- b) (i) For a complex number z , where n is a positive integer and $|z| = 1$, show that 1
- $$z^n + \bar{z}^n = 2 \cos n\theta.$$
- (ii) Given that $\theta \neq (2k+1)\pi$, where $k \in \mathbb{Z}$, prove that $\frac{e^{i\theta} - 1}{e^{i\theta} + 1} = i \tan\left(\frac{\theta}{2}\right)$. 2
- c) Let z_1, z_2, z_3 and z_4 are four complex numbers such that:
- $$|z_1| = |z_2| = |z_3| = 1, \quad |z_4| = 3 \quad \text{and} \quad z_1 + z_2 + z_3 = 0$$
- (i) Show that $|z_4 - z_1|^2 = 10 - (z_4 \bar{z}_1 + \bar{z}_4 z_1)$. 2
- (ii) Hence show that $|z_4 - z_1|^2 + |z_4 - z_2|^2 + |z_4 - z_3|^2 = 30$. 2

END OF EXAMINATION

Year-12 Assessment Task-1, 2025 Mathematics Extension-2

Marking Guidelines

Section - I

Multiple Choice (1 - 5)

1) C

2) A

3) B

4) C

5) A

Sample Solutions:

$$1) i = e^{i\pi/2}$$

$$i^3 = e^{i3\pi/2}$$

$$\sqrt{i^3} = e^{i3\pi/4} \quad (C)$$

2) $3+i$ and $3-i$ are roots

$$\text{Sum of roots} = 6$$

$$\text{Product of roots} = (3+i)(3-i) = 10$$

$\therefore z^2 - 6z + 10$ is a factor of $P(z)$. (A)

$$3) \text{ Let } z = r e^{i\theta}$$

$$w = \frac{z^2}{i|z|} = \frac{r^2 e^{i2\theta}}{r \cdot e^{i\pi/2}} = r e^{i(2\theta - \pi/2)}$$

$\therefore w$ has same modulus as z . (B)

$\arg w$ is in first quadrant. ($\because \theta$ is in first quadrant, 2θ is in second quadrant, $(2\theta - \frac{\pi}{2})$ will be in first quadrant)

$$\begin{aligned} 4) (-1+i)^n &= \left(\sqrt{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right) \right)^n \\ &= \sqrt{2}^n \left(\cos \frac{3n\pi}{4} + i \sin \frac{3n\pi}{4} \right) \quad (C) \end{aligned}$$

$$5) \text{ Since } z^2 + z + 1 = 0 \Rightarrow \frac{z^3 - 1}{z - 1} = 0, z \neq 1$$

$\therefore z = \omega, \omega^2$, where ω is a cube root of unity.

$$(\text{Also } \omega^3 = 1, \frac{1}{\omega} = \omega^2 \text{ or } \frac{1}{\omega^2} = \omega)$$

$$\text{Let } z = \omega$$

$$\begin{aligned} & \therefore \left(z + \frac{1}{z}\right)^2 + \left(z^2 + \frac{1}{z^2}\right)^2 + \dots + \left(z^6 + \frac{1}{z^6}\right)^2 \\ &= \left(\omega + \frac{1}{\omega}\right)^2 + \left(\omega^2 + \frac{1}{\omega^2}\right)^2 + \left(\omega^3 + \frac{1}{\omega^3}\right)^2 + \left(\omega^4 + \frac{1}{\omega^4}\right)^2 + \left(\omega^5 + \frac{1}{\omega^5}\right)^2 \\ & \quad + \left(\omega^6 + \frac{1}{\omega^6}\right)^2 \\ &= (\omega + \omega^2)^2 + (\omega^2 + \omega)^2 + (1 + 1)^2 + (\omega + \omega^2)^2 + (\omega^2 + \omega)^2 \\ & \quad + (1 + 1)^2 \\ &= 4(\omega + \omega^2)^2 + 4 + 4 \\ &= 4(-1)^2 + 4 + 4 \\ &= 12 \end{aligned} \quad (A)$$

END OF SECTION-I

Section II

Question-6

a) i) $w = 1 + iz$

1 mark Correct Solution

$$w = 1 + i(2 + i)$$

$$= 1 + 2i + i^2$$

$$= 2i$$

ii) $w = 2 - i$, $z = ?$

1 mark Correct Solution

$$2 - i = 1 + iz$$

$$iz = 1 - i$$

$$z = \frac{1}{i} - 1$$

$$= -1 - i$$

b) $z = (b + i)^2$

3 marks Correct Solution.

$$= b^2 + i2b - 1$$

2 marks Solving the quadratic equation.

$$= (b^2 - 1) + i2b$$

Since $\arg z = \frac{\pi}{3}$

1 mark Finding the quadratic equation by using $\arg$. or equivalent

$$\therefore \frac{2b}{b^2 - 1} = \tan \frac{\pi}{3}$$

$$\frac{2b}{b^2 - 1} = \sqrt{3}$$

$$\sqrt{3}b^2 - 2b - \sqrt{3} = 0$$

$$(\sqrt{3}b + 1)(b - \sqrt{3}) = 0$$

$$b = \sqrt{3}, -\frac{1}{\sqrt{3}}$$

$$b = \sqrt{3} \quad (\because b > 0 \text{ as } \arg z \text{ is acute})$$

$$c) z^2 - 5z + (7+i) = 0$$

$$\Delta = (-5)^2 - 4(7+i)$$

$$= 25 - 28 - 4i$$

$$= -3 - 4i$$

$$\sqrt{-3-4i} = a+ib, \quad a, b \in \mathbb{R}$$

$$-3-4i = a^2 - b^2 + 2iab$$

$$a^2 - b^2 = -3, \quad ab = -2$$

$$\text{Solve then } a+ib = \pm(1-2i)$$

$$\therefore \sqrt{-3-4i} = \pm(1-2i)$$

$$z = \frac{5 \pm \sqrt{-3-4i}}{2}$$

$$= \frac{5 \pm (1-2i)}{2}$$

$$= \frac{6-2i}{2}, \quad \frac{4+2i}{2}$$

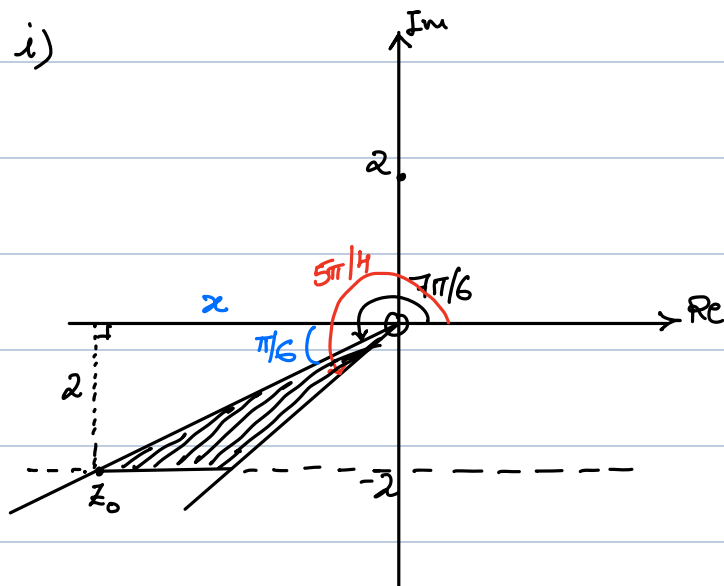
$$= 3-i, \quad 2+i$$

3 marks Correct Solution.

2 marks Finding the square root of discriminant.

1 mark Finding the discriminant of given equation.

d) i)



2 marks Correct solution

including region.

1 mark Correct graph of any one inequality.

$$ii) \tan \frac{\pi}{6} = \frac{2}{x}$$

$$\frac{1}{\sqrt{3}} = \frac{2}{x} \Rightarrow x = 2\sqrt{3} \quad \therefore z_0 = -2\sqrt{3} - 2i$$

1 mark Correct solution

Question: 7

Since $(a+ib)$ and $(b+ia)$ are roots

$\therefore (a-ib)$ and $(b-ia)$ are also roots

Sum of the roots = -4

$$\therefore a+ib + a-ib + b+ia + b-ia = -4$$

$$\therefore 2a + 2b = -4$$

$$a + b = -2 \quad \text{--- (1)}$$

Product of the roots = 400

$$\therefore (a+ib)(a-ib)(b+ia)(b-ia) = 400$$

$$(a^2+b^2)(a^2+b^2) = 400$$

$$(a^2+b^2)^2 = 400$$

$$a^2+b^2 = 20 \quad (\because a^2+b^2 > 0)$$

from (1) $b = -2-a$

$$\therefore a^2 + (-2-a)^2 = 20$$

$$a^2 + a^2 + 4a + 4 = 20$$

$$2a^2 + 4a - 16 = 0$$

$$a^2 + 2a - 8 = 0$$

$$(a+4)(a-2) = 0$$

$$a = -4, 2$$

3 marks Correct solution

2 marks Using sum and

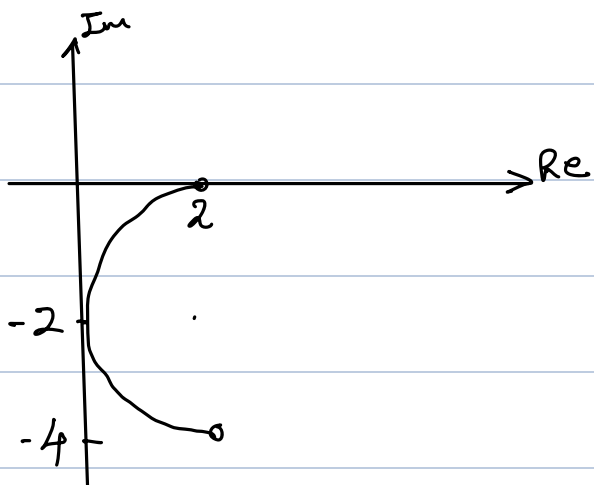
Product of roots to

find the equations in terms of a & b or equivalent merit

1 mark Finding the conjugate

of the given roots or four roots of given eq.

b)



2 marks correct solution

1 mark sketching the locus
as semi-circle in
any form.

c) Geometric Approach:

$$\text{Let } w = \frac{z-1}{z+1}$$

$$wz + w = z - 1$$

$$w + 1 = z - wz$$

$$w + 1 = z(1 - w)$$

$$z = \frac{w+1}{1-w}$$

$$\text{Since } |z| = 1$$

$$\therefore \left| \frac{w+1}{1-w} \right| = 1$$

$$\therefore |w+1| = |w-1|$$

which represents the distance of complex number w from 1 and -1 is same. That means w lies on y -axis or imaginary axis of Argand diagram.

Therefore $w = \frac{z-1}{z+1}$ is purely imaginary number.

OR

2 marks correct proof

1 mark using $|z| = 1$ and
rearranging $z = \frac{w+1}{w-1}$

Algebraic Approach:

$$\text{Let } z = a + ib, \quad a, b \in \mathbb{R}$$

$$|z| = 1 \Rightarrow a^2 + b^2 = 1$$

2 marks Correct proof

$$\frac{z-1}{z+1} = \frac{a+ib-1}{a+ib+1}$$

1 mark Express $\frac{z-1}{z+1}$ in

simplest form by

assuming either

$z = a+ib$ or $|z| = 1$

$$= \frac{(a-1)+ib}{(a+1)+ib} \times \frac{(a+1)-ib}{(a+1)-ib}$$

$$= \frac{(a^2-1)+b^2 + i[b(a+1)-b(a-1)]}{(a+1)^2 + b^2}$$

$$= \frac{(a^2+b^2-1) + 2ib}{(a+1)^2 + b^2}$$

$$= \frac{2ib}{(a+1)^2 + b^2} \quad (\because a^2+b^2=1)$$

$\therefore \frac{z-1}{z+1}$ is purely imaginary.

$$\text{d) i) R.H.S} = (z+1)(z^4 - z^3 + z^2 - z + 1)$$

$$= z(z^4 - z^3 + z^2 - z + 1) + (z^4 - z^3 + z^2 - z + 1)$$

$$= z^5 - z^4 + z^3 - z^2 + z + z^4 - z^3 + z^2 - z + 1$$

$$= z^2 + 1$$

$$= \text{L.H.S}$$

1 mark Correct proof.

ii) Since z is complex root of $z^5 + 1 = 0$

$$\therefore (z+1)(z^4 - z^3 + z^2 - z + 1) = 0$$

$$\Rightarrow z^4 - z^3 + z^2 - z + 1 = 0 \quad (\because z \neq -1)$$

$$\Rightarrow z^4 + z^2 + 1 = z^3 + z$$

Hence proved.

1 mark Correct proof.

iii) If $z^5 = -1$

then $z = \text{Cis}\left(\pm \frac{\pi}{5}\right), \text{Cis}\left(\pm \frac{3\pi}{5}\right), 1$

Let $z = \text{Cis} \frac{\pi}{5}$

$$1 + z^2 + z^4 = z + z^3 \quad (\text{from (ii)})$$

$$(\div z^2) \quad \frac{1}{z^2} + 1 + z^2 = \frac{1}{z} + z$$

$$\left(\frac{1}{z^2} + z^2\right) + 1 = \left(\frac{1}{z} + z\right)$$

$$2 \cos \frac{2\pi}{5} + 1 = 2 \cos \frac{\pi}{5}$$

$$-2 \cos \frac{3\pi}{5} + 1 = 2 \cos \frac{\pi}{5}$$

$$2 \cos \frac{\pi}{5} + 2 \cos \frac{3\pi}{5} = 1$$

$$\cos \frac{\pi}{5} + \cos \frac{3\pi}{5} = \frac{1}{2}$$

Hence Shown

3 marks Correct solution.

2 marks Using part (ii) to express it in $\cos \frac{n\pi}{5}$ form or equivalent merit.

1 mark Listing all the five roots of z .

Question-8

a) i) $|w|=1$ and $\arg w = \frac{\pi}{3}$ (60°)

$\therefore |zw|=|z|$ and $\arg(zw) = \arg z + \frac{\pi}{3}$

$\therefore \triangle OAB$ is equilateral. (2 sides are equal and included angle is 60°)

2 marks Correct solution.

1 mark Either show $|zw|=|z|$ or $|w|=1$ and its $\arg \frac{\pi}{3}$.

ii) $\overrightarrow{BA} = z - zw$

$\overrightarrow{OD} = \overrightarrow{BA}$

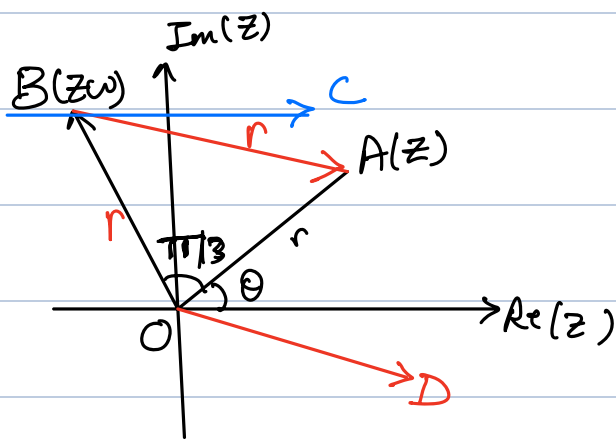
$= z - zw$

$\overrightarrow{OD} = \overrightarrow{OA} \cdot e^{-i\pi/3}$

$= z \cdot e^{-i\pi/3}$

$= re^{i\theta} \cdot e^{-i\pi/3}$

$= re^{i(\theta - \pi/3)}$



2 marks Correct solutions

1 mark Attempt to find the argument of $(z - zw)$ or equivalent merit.

$$b) \text{ L.H.S} = \frac{e^{i\theta} - 1}{e^{i\theta} + 1}$$

$$= \frac{(e^{i\theta} - 1) \div e^{i\theta/2}}{(e^{i\theta} + 1) \div e^{i\theta/2}}$$

$$= \frac{e^{i\theta/2} - e^{-i\theta/2}}{e^{i\theta/2} + e^{-i\theta/2}}$$

$$= \frac{2i \sin \theta/2}{2 \cos \theta/2}$$

$$= i \tan \theta/2$$

$$= \text{R.H.S}$$

OR

$$\frac{e^{i\theta} - 1}{e^{i\theta} + 1} = \frac{(\cos \theta - 1) + i \sin \theta}{(\cos \theta + 1) + i \sin \theta}$$

$$= \frac{(\cos \theta - 1) + i \sin \theta}{(\cos \theta + 1) + i \sin \theta} \times \frac{(\cos \theta + 1) - i \sin \theta}{(\cos \theta + 1) - i \sin \theta}$$

$$= \frac{(\cos^2 \theta - 1) + \sin^2 \theta + i[\sin \theta(\cos \theta + 1) - \sin \theta(\cos \theta - 1)]}{(\cos \theta + 1)^2 + \sin^2 \theta}$$

$$= \frac{2i \sin \theta}{\cos^2 \theta + 2\cos \theta + 1 + \sin^2 \theta}$$

$$= \frac{2i \sin \theta}{2(1 + \cos \theta)} = \frac{2i \sin \theta/2 \cos \theta/2}{2 \cos^2 \theta/2}$$

$$= i \tan \theta/2$$

3 marks Correct Solution

2 marks Express $e^{i\theta}$ as $\cos \theta$ to

rationalise & simplifying
to $\frac{i \sin \theta}{1 + \cos \theta}$ or in terms
of $\frac{e^{i\theta/2} - e^{-i\theta/2}}{e^{i\theta/2} + e^{-i\theta/2}}$

1 mark Expressing $e^{i\theta}$ as $\cos \theta$

and rationalising den.

or dividing Num and
den. by $e^{i\theta/2}$

$$\begin{aligned}
\text{C) i) L.H.S} &= |z - z_1|^2 \\
&= (z - z_1)(\overline{z - z_1}) \quad (\because |z|^2 = z\bar{z}) \\
&= (z - z_1)(\bar{z} - \bar{z}_1) \\
&= z\bar{z} - z\bar{z}_1 - z_1\bar{z} + z_1\bar{z}_1 \\
&= |z|^2 - (z\bar{z}_1 + z_1\bar{z}) + |z_1|^2 \\
&= 9 - (z\bar{z}_1 + z_1\bar{z}) + 1 \\
&= 10 - (z\bar{z}_1 + z_1\bar{z})
\end{aligned}$$

2 marks correct solution

1 mark Expressing $|z|^2 = z\bar{z}$ and expanding or equivalent merit.

$$\text{ii) } |z - z_1|^2 = 10 - (z\bar{z}_1 + \bar{z}z_1) \quad (\text{using (i)})$$

$$\text{Similarly } |z - z_2|^2 = 10 - (z\bar{z}_2 + \bar{z}z_2)$$

$$\& \quad |z - z_3|^2 = 10 - (z\bar{z}_3 + \bar{z}z_3)$$

Adding them.

$$|z - z_1|^2 + |z - z_2|^2 + |z - z_3|^2 =$$

$$= 30 - [z(\bar{z}_1 + \bar{z}_2 + \bar{z}_3) + \bar{z}(z_1 + z_2 + z_3)]$$

$$= 30 - [z(\overline{z_1 + z_2 + z_3}) + \bar{z}(0)]$$

$$= 30 - [z(0) + 0]$$

$$= 30$$

2 marks correct solution

1 mark Using part (i) to express the L.H.S in terms of $z, \bar{z}, z_1, z_2$ & z_3 .

END OF EXAM